REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	4333
Normal Flows	4060
Detected Attacks	273
Attack Rate	6.30%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.30
Validation Accuracy	80.96%
Attack Precision	0.93
Attack Recall	0.72
Attack F1-Score	0.81

3. Network Protocol Distribution

Protocol	Flow Count
TCP	2508
UDP	1823
OTHER	2

4. Detected Attack History

Time	Source	Destination	Protocol	Confidence
2026-08-25 17:49:05	10.2.0.1:65432	192.168.1.18:58178	TCP	42.48%
2026-08-25 17:48:55	172.64.155.209:443	192.168.1.18:49807	TCP	31.99%
2026-08-25 17:48:38	142.250.4.101:443	192.168.1.18:51144	UDP	61.74%
2026-08-25 17:48:04	192.168.1.18:62659	20.42.73.26:443	TCP	45.43%
2026-08-25 17:48:04	10.2.0.1:65432	192.168.1.18:52706	TCP	42.68%
2026-08-25 17:47:50	192.168.1.18:62659	20.42.73.26:443	TCP	51.47%
2026-08-25 17:47:01	10.2.0.1:65432	192.168.1.18:55330	TCP	42.68%
2026-08-25 17:46:44	192.168.1.18:64631	202.88.211.108:443	UDP	30.55%
2026-08-25 17:46:43	150.171.110.83:443	192.168.1.18:55329	TCP	47.98%
2026-08-25 17:46:22	192.168.1.18:64631	202.88.211.108:443	UDP	30.07%
2026-08-25 17:46:16	172.64.155.209:443	192.168.1.18:49807	TCP	35.88%
2026-08-25 17:45:58	192.168.1.18:64631	202.88.211.108:443	UDP	30.07%
2026-08-25 17:45:53	10.2.0.1:65432	192.168.1.18:51403	TCP	42.68%
2026-08-25 17:44:46	10.2.0.1:65432	192.168.1.18:51399	TCP	42.68%
2026-08-25 17:43:44	192.168.1.18:62661	216.137.52.52:443	TCP	30.18%
2026-08-25 17:43:37	10.2.0.1:65432	192.168.1.18:58197	TCP	42.68%
2026-08-25 17:42:55	192.168.1.18:62659	20.42.73.26:443	TCP	49.18%
2026-08-25 17:42:53	192.168.1.18:58189	20.250.119.64:443	TCP	32.32%
2026-08-25 17:42:49	192.168.1.18:61472	4.207.44.66:443	TCP	30.73%
2026-08-25 17:42:40	172.64.155.209:443	192.168.1.18:49807	TCP	32.54%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.